Experiment 02: Agile Software Development Lifecycle using Jira

Experiment No.	01
Mapped CO	CO2
Aim	To study and implement the Agile Software Development Lifecycle using Jira by creating and managing a Scrum project, including product backlog, user stories, sprint planning, task assignment, and sprint tracking.
Tools / Software	Git (>= 2.30); A GitHub account; A code editor / terminal

Theory

Agile is an iterative software development methodology that focuses on delivering software in small, incremental releases while continuously incorporating customer feedback. It promotes collaboration, flexibility, and rapid delivery of high-quality software.

Scrum is one of the most widely used Agile frameworks. It organizes work into fixed-duration iterations called **Sprints**, during which the development team completes selected tasks from the **Product Backlog**. Scrum defines roles such as **Product Owner**, **Scrum Master**, and **Development Team**, and includes events like Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective.

Jira, developed by Atlassian, is a popular Agile project management tool that enables teams to create projects, manage user stories, plan sprints, assign work, track progress using Scrum boards, and generate reports such as Burndown Charts and Sprint Reports.

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Procedure

- Create a Jira Cloud account and log in.
- Create a new Scrum Software Project.
- Enter the project name and create the project.
- Create Epics for different functional modules.
- Create User Stories using the format **As a..., I want..., So that...**
- Add all stories to the Product Backlog.
- Create Sprint 1.
- Move selected User Stories into the Sprint.

- Start the Sprint by specifying its duration and Sprint Goal.
- Create Tasks/Subtasks for each User Story.
- Assign tasks to team members.
- Update task status from **To Do → In Progress → Done**.
- View Scrum Board to monitor project progress.
- Generate Burndown Chart and Sprint Report.

Sample Code / Commands

Sample Activity

Project Name

Online Food Ordering System

Example User Story

Story: User Registration

As a new customer,

I want to register on the application,

So that I can order food online.

Example Sprint

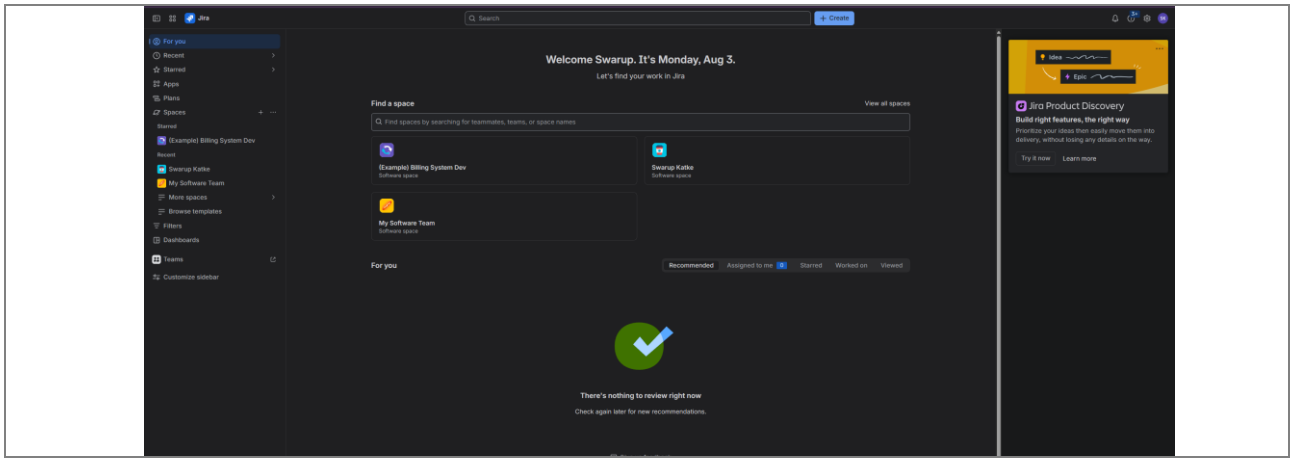
Sprint Name : Sprint 1

Duration : 2 Weeks

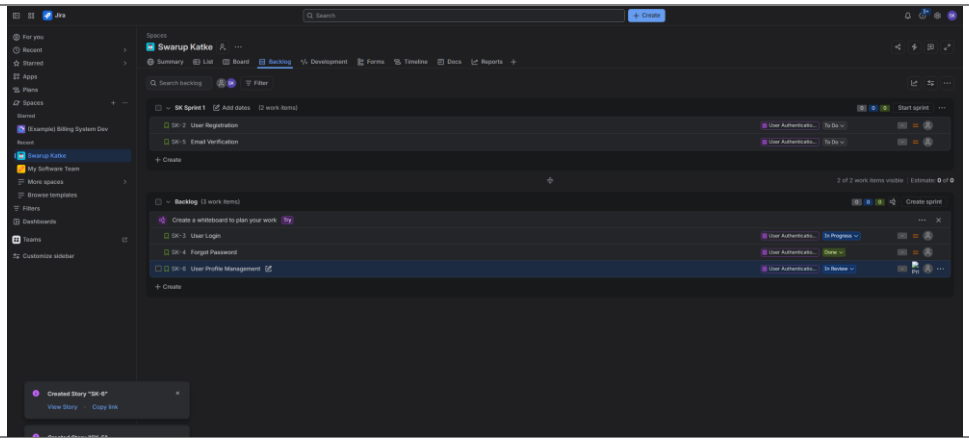
Sprint Goal :

Complete User Authentication Module

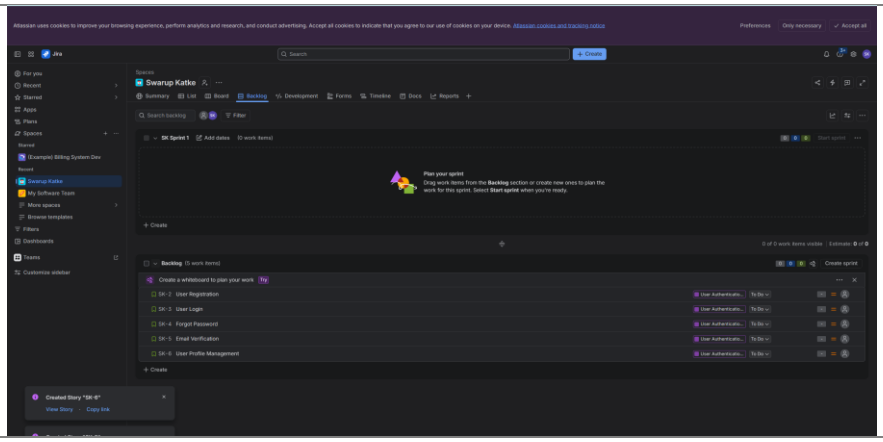
1. **Jira Dashboard**



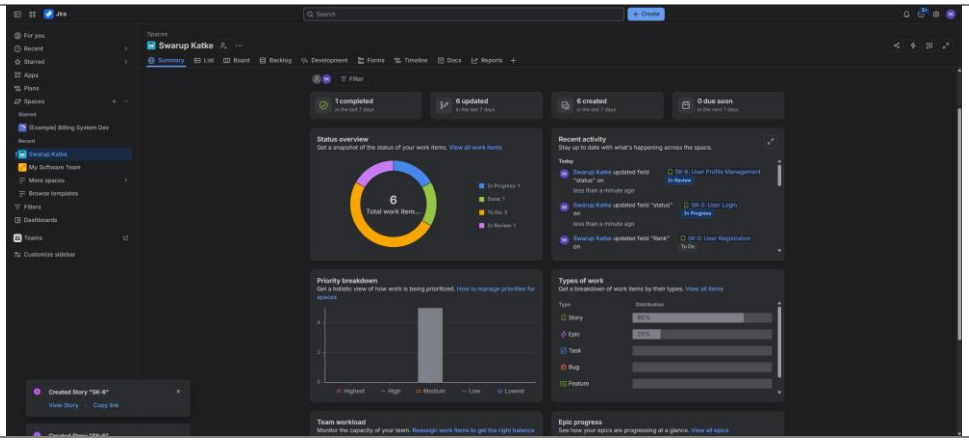
2. Product Backlog



3. Sprint Planning



4. Scrum Board



Conclusion: Successfully implemented the Agile Software Development Lifecycle using Jira by creating and managing a Scrum project. They learned to organize work using Epics, User Stories, Product Backlog, and Sprints, assign tasks to team members, monitor project progress through the Scrum Board, and analyze project performance using Jira reports. This experiment provided practical exposure to Agile project management and collaborative software development.